

Reconstruction Metric - GPU Optimized

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```
# Configuración para visualización en PDF
import pandas as pd
import numpy as np
import sys
import os
from sae.tools.experiment_utils import get_metrics_dir

pd.set_option('display.max_colwidth', None)
pd.set_option('display.max_columns', None)
pd.set_option('display.width', 100)
np.set_printoptions(linewidth=100, edgeitems=3)

os.environ['COLUMNS'] = '100'

print(" Configuración para PDF lista")
```

Configuración para PDF lista

```
NAMING_META = {
    'variante': 'standard',          # Arquitectura del SAE
    'tecnica': 'p-annealing',        # P-annealing (Lp^p con p decreciente)
    'capa': 6,                       # Layer del modelo OthelloGPT
    'juegos': 12500,                 # Número de partidas
    'base_path': 'C:\\Users\\Esposa\\Documents\\Repos\\sae-othello-gpt', # Path base para checkpoints
    'experiment_base_path': os.path.abspath('../../experiments'),      # sae/experiments/
    'extras': {
        'expansion': 16,
        'sparsity_coef': 3e-1,
        'epochs': 200,
        'early_stopping': False,
        'learning_rate': 1e-3,
        'p_start': 1.0,
        'p_end': 0.2,
        'n_sparsity_updates': 10,
        'anneal_start': 5000,
        'batch_size': 2048,
        'warmup_steps': 1000,
        'sparsity_warmup_steps': 2000,
```

```

        'estimated_epochs': 200
    }
}

print('\n Metadata de naming:')
for key, value in NAMING_META.items():
    print(f' {key}: {value}')
```

```

Metadata de naming:
variante: standard
tecnica: p-annealing
capa: 6
juegos: 12500
base_path: C:\Users\Esposa\Documents\Repos\sae-othello-gpt
experiment_base_path: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments
extras: {'expansion': 16, 'sparsity_coef': 0.3, 'epochs': 200, 'early_stopping': False, 'learning_rate': 0.001, 'p_st
```

0.1 1. Setup e Imports

```

import numpy as np
import torch
import sys
import time
import matplotlib.pyplot as plt

from sae.models.sae import SparseAutoencoder
from sae.tools.naming_utils import get_checkpoint_dir, get_model_name, get_extras_id
from pathlib import Path
from tqdm import tqdm

project_root = Path('../..').resolve()
sys.path.insert(0, str(project_root))

device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
print(f"Device: {device}")
if torch.cuda.is_available():
    print(f"GPU: {torch.cuda.get_device_name(0)}")
    print(f"Memoria: {torch.cuda.get_device_properties(0).total_memory / 1e9:.2f} GB")
```

```

Device: cuda
GPU: NVIDIA GeForce RTX 5060
Memoria: 8.55 GB
```

0.2 2. Cargar Datos

```

# Cargar activaciones
activations_path = project_root / "activations" / "data" / "layer6_2000games.npy"
activations = np.load(activations_path)

print(f"Activaciones del modelo:")
print(f" Shape: {activations.shape}")
print(f" Memoria: {activations.nbytes / (1024**2):.2f} MB")
```

```

Activaciones del modelo:
Shape: (118000, 512)
Memoria: 230.47 MB
```

```
# Cargar ground truth de BSPs
bsp_gt_path = project_root / "metrics" / "02_data" / "bsp_ground_truth_2000games.npy"
bsp_names_path = project_root / "metrics" / "02_data" / "bsp_ground_truth_2000games.names.npy"

bsp_ground_truth = np.load(bsp_gt_path)
bsp_names = np.load(bsp_names_path, allow_pickle=True)

print(f"\nGround truth BSPs:")
print(f"  Shape: {bsp_ground_truth.shape}")
print(f"  Total BSPs: {len(bsp_names)}")
```

```
Ground truth BSPs:
  Shape: (118000, 326)
  Total BSPs: 326
```

0.3 3. Cargar SAE y Extraer Features

```
input_dim = 512
hidden_dim = 8192

sae = SparseAutoencoder(input_dim, hidden_dim).to(device)
checkpoint_dir = get_checkpoint_dir(
    NAMING_META['base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos']
)
extras_id = get_extras_id(checkpoint_dir, NAMING_META['extras']) if NAMING_META.get('extras') else None

model_filename = get_model_name(
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    'final',
    extras_id=extras_id
)
model_path = checkpoint_dir / model_filename
checkpoint = torch.load(model_path, map_location=device, weights_only=False)
sae.load_state_dict(checkpoint['model_state_dict'])
sae.eval()

# Extraer info de epoch/val_mse según formato del checkpoint
metadata = checkpoint.get('metadata', {})
history = checkpoint.get('history', {})
epoch = checkpoint.get('epoch', history.get('best_epoch', metadata.get('estado', 'N/A')))
val_mse = checkpoint.get('val_mse', history.get('best_val_mse', 'N/A'))

print(f"SAE cargado:")
print(f"  Path: {model_path}")
print(f"  Input: {input_dim}, Hidden: {hidden_dim}")
print(f"  Expansion: {hidden_dim/input_dim}x")
print(f"  Epoch: {epoch}, Val MSE: {val_mse if val_mse == 'N/A' else f'{val_mse:.6f}'}")
print(f"  Extras ID: {extras_id}")
print(f"  Claves checkpoint: {list(checkpoint.keys())}")
```

```
SAE cargado:
```

```

Path: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\sae_p-annealing_standard\12500game
annealing_16_12500g_ex3_final.pt
Input: 512, Hidden: 8192
Expansion: 16.0x
Epoch: final, Val MSE: N/A
Extras ID: ex3
Claves checkpoint: ['model_state_dict', 'config', 'metadata', 'history', 'optimizer_state_dict']

```

```

# Extraer features del SAE en GPU
print("Extrayendo features del SAE...")
activations_tensor = torch.from_numpy(activations).float().to(device)

with torch.no_grad():
    sae_features = torch.relu(sae.encoder(activations_tensor))

print(f"\nFeatures SAE:")
print(f"  Shape: {sae_features.shape}")
print(f"  Device: {sae_features.device}")
sparsity = (sae_features == 0).float().mean().item()
print(f"  Sparsity: {sparsity:.2%}")
print(f"  Activaciones promedio: {sae_features.shape[1] * (1 - sparsity):.1f}")

```

Extrayendo features del SAE...

```

Features SAE:
  Shape: torch.Size([118000, 8192])
  Device: cuda:0
  Sparsity: 85.22%
  Activaciones promedio: 1210.5

```

```

del activations_tensor
torch.cuda.empty_cache()

```

0.4 4. Split Train/Test y Filtrar BSPs de Piezas

```

# Split 50/50:
n_moves = 59
n_games = 2000 # número de partidas válidas
train_games = n_games // 2
split_idx = train_games * n_moves

sae_features_train = sae_features[:split_idx]
sae_features_test = sae_features[split_idx:]

print(f"Total partidas: {n_games}")
print(f"Train: {train_games} partidas ({split_idx} activaciones)")
print(f"Test: {n_games - train_games} partidas ({len(activations) - split_idx} activaciones)")
print(f"Train set: {sae_features_train.shape}")
print(f"Test set: {sae_features_test.shape}")

```

```

Total partidas: 2000
Train: 1000 partidas (59000 activaciones)
Test: 1000 partidas (59000 activaciones)
Train set: torch.Size([59000, 8192])
Test set: torch.Size([59000, 8192])

```

```

# =====
# CONFIGURACIÓN DE FILTRADO DE BSPs
# =====

```

```

USE_PLAYER    = False    # BSPs perspectiva del jugador (1 y 2)
USE_ABSOLUTE  = True     # BSPs absolutas negro/blanco (B y W)
USE_STRATEGIC = False    # BSPs especiales (BSP_)

bsp_pieces_indices = []
bsp_pieces_names = []

for i, name in enumerate(bsp_names):
    include = False
    if USE_PLAYER and len(name) == 6 and name.startswith('BSP') and (name.endswith('1') or name.endswith('2')):
        include = True
    if USE_ABSOLUTE and (name.endswith('B') or name.endswith('W')):
        include = True
    if USE_STRATEGIC and name.startswith('BSP_'):
        include = True
    if include:
        bsp_pieces_indices.append(i)
        bsp_pieces_names.append(name)

bsp_pieces_indices = np.array(bsp_pieces_indices)

# Para reconstruction necesitas split train/test
bsp_gt_train_pieces = torch.from_numpy(bsp_ground_truth[:split_idx, bsp_pieces_indices]).bool().to(device)
bsp_gt_test_pieces  = torch.from_numpy(bsp_ground_truth[split_idx:, bsp_pieces_indices]).bool().to(device)

parts = []
if USE_PLAYER: parts.append("player")
if USE_ABSOLUTE: parts.append("absolute")
if USE_STRATEGIC: parts.append("strategic")
filter_suffix = "_".join(parts) if parts else "none"

print(f"BSPs seleccionadas para Reconstruction:")
print(f"  Player: {USE_PLAYER} | Absolute: {USE_ABSOLUTE} | Strategic: {USE_STRATEGIC}")
print(f"  Total: {len(bsp_pieces_indices)}")
print(f"  Train shape: {bsp_gt_train_pieces.shape}")
print(f"  Test shape: {bsp_gt_test_pieces.shape}")
print(f"  Device: {bsp_gt_train_pieces.device}")
print(f"  Primeras 5: {'', '.join(bsp_pieces_names[:5])}")
print(f"  Últimas 5: {'', '.join(bsp_pieces_names[-5:])}")
print(f"  Filter suffix: {filter_suffix}")

```

```

BSPs seleccionadas para Reconstruction:
  Player: False | Absolute: True | Strategic: False
  Total: 128
  Train shape: torch.Size([59000, 128])
  Test shape: torch.Size([59000, 128])
  Device: cuda:0
  Primeras 5: BSPA1B, BSPA1W, BSPA2B, BSPA2W, BSPA3B
  Últimas 5: BSPH6W, BSPH7B, BSPH7W, BSPH8B, BSPH8W
  Filter suffix: absolute

```

0.5 5. Funciones GPU-Optimizadas

```

def fast_precision_score_gpu(y_true, y_pred, eps=1e-8):
    """
    Precisión vectorizada en GPU.

    Args:
        y_true: Tensor (n_positions,) booleano
        y_pred: Tensor (n_positions, n_candidates) booleano
    """

```

```

Returns:
    precision_scores: Tensor (n_candidates,)
    """
    y_true_expanded = y_true.unsqueeze(1)

    tp = (y_true_expanded & y_pred).sum(dim=0).float()
    fp = (~y_true_expanded & y_pred).sum(dim=0).float()

    precision = tp / (tp + fp + eps)

    return precision

def fast_f1_score_gpu(y_true, y_pred, eps=1e-8):
    """
    F1 score vectorizado en GPU.

    Args:
        y_true: Tensor (n_positions,) booleano
        y_pred: Tensor (n_positions,) booleano

    Returns:
        f1: Float
    """
    tp = (y_true & y_pred).sum().float()
    fp = (~y_true & y_pred).sum().float()
    fn = (y_true & ~y_pred).sum().float()

    precision = tp / (tp + fp + eps)
    recall = tp / (tp + fn + eps)

    f1 = 2 * (precision * recall) / (precision + recall + eps)

    return f1.item()

print(" Funciones GPU definidas")

```

Funciones GPU definidas

0.6 6. Fase 1: Identificar Features de Alta Precisión (GPU)

Para cada BSP, encontrar features con precisión 0.95 en el train set.

```

def identify_high_precision_features_gpu(
    sae_features_train,
    bsp_gt_train,
    f_max,
    precision_threshold=0.95,
    significance_threshold=10,
    thresholds=[0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9],
    feature_batch_size=512,
    verbose=True
):
    """
    Para cada (threshold t, feature f, BSP b): determina si f es clasificador
    de alta precisión para b al threshold t.

    Criterios (igual que implementación oficial):
        - precision(f, b, t) >= precision_threshold (default 0.95)
        - on_count(f, t) >= significance_threshold (default 10)
    """

```

```

Args:
    sae_features_train: Tensor (P, n_features) GPU
    bsp_gt_train: Tensor (P, n_bsps) GPU, bool
    f_max: Tensor (n_features,) GPU - calculado FUERA para
            garantizar identidad exacta con Fase 2
    precision_threshold: float, default 0.95
    significance_threshold: int, mín. activaciones requeridas (default 10)
    thresholds: lista de fracciones t [0, 1)
    feature_batch_size: int, features por mini-batch interno

Returns:
    hp_mask_TFB: BoolTensor (T, n_active, n_bsps)
    active_feature_indices: LongTensor (n_active,)
"""
n_positions, n_features = sae_features_train.shape
n_bsps = bsp_gt_train.shape[1]
device = sae_features_train.device

thresholds_T = torch.tensor(thresholds, device=device)
n_thresholds = len(thresholds_T)

# Features vivas (f_max > 0)
active_feature_indices = (f_max > 0).nonzero(as_tuple=True)[0]
n_active = len(active_feature_indices)

if verbose:
    print("=" * 60)
    print("FASE 1: IDENTIFICAR FEATURES DE ALTA PRECISIÓN")
    print("=" * 60)
    print(f"Precision threshold: {precision_threshold}")
    print(f"Significance threshold: >= {significance_threshold} activaciones")
    print(f"Features activas: {n_active}/{n_features} ({n_active/n_features*100:.1f}%)" )
    print(f"BSPs: {n_bsps}")
    print(f"Thresholds: {thresholds}")
    print("=" * 60)

# Thresholds absolutos: thresh_TF[t, f] = t * f_max[f]
active_f_max = f_max[active_feature_indices] # (n_active,)
thresholds_TF = thresholds_T[:, None] * active_f_max # (T, n_active)

# Acumuladores sobre todo el train set
on_count_TF = torch.zeros(n_thresholds, n_active, device=device)
tp_count_TFB = torch.zeros(n_thresholds, n_active, n_bsps, device=device)

bsp_float_PB = bsp_gt_train.float() # (P, B)

n_iters = (n_active + feature_batch_size - 1) // feature_batch_size

for fi in tqdm(range(n_iters), desc="Fase 1"):
    f_start = fi * feature_batch_size
    f_end = min(f_start + feature_batch_size, n_active)

    global_idx = active_feature_indices[f_start:f_end]
    acts_PF = sae_features_train[:, global_idx] # (P, bs)
    thresh_TF = thresholds_TF[:, f_start:f_end] # (T, bs)

    # active_TFP[t, f, p] = True si acts[p,f] > thresh[t,f]
    active_TFP = acts_PF.T.unsqueeze(0) > thresh_TF.unsqueeze(2) # (T, bs, P)

    # Conteo de activaciones por (threshold, feature)
    on_count_TF[:, f_start:f_end] += active_TFP.sum(dim=2).float()

```

```

# True positives: f activa Y BSP verdadera
# tp[t, f, b] =  $\Sigma_p$  active[t,f,p] * bsp[p,b]
tp_count_TFB[:, f_start:f_end, :] += torch.einsum(
    'tfp,pb->tfb', active_TFP.float(), bsp_float_PB
)

# Precisión: tp / on_count
eps = 1e-8
precision_TFB = tp_count_TFB / (on_count_TF.unsqueeze(2) + eps) # (T, F, B)

# Máscara: alta precisión AND significancia suficiente
sig_mask_TF1 = (on_count_TF >= significance_threshold).unsqueeze(2) # (T, F, 1)
hp_mask_TFB = (precision_TFB >= precision_threshold) & sig_mask_TF1 # (T, F, B)

if verbose:
    hp_per_t = hp_mask_TFB.sum(dim=(1, 2)) # (T,)
    hp_per_b = hp_mask_TFB.sum(dim=1) # (T, B)
    best_t = hp_per_t.argmax().item()
    print(f"\nPares (feature, BSP) de alta precisión por threshold:")
    for i, t in enumerate(thresholds):
        marker = " ← más pares" if i == best_t else ""
        print(f" t={t:.1f}: {hp_per_t[i].item():5d} pares{marker}")
    print(f"\nFeatures HP por BSP en t={thresholds[best_t]:.1f}:")
    print(f" Media: {hp_per_b[best_t].float().mean():.1f}")
    print(f" Máx: {hp_per_b[best_t].max().item()}")
    print(f" BSPs sin features: {(hp_per_b[best_t] == 0).sum().item()}")

return hp_mask_TFB, active_feature_indices

print(" Fase 1 rediseñada: umbral de significancia (10) + estructura (T, F, B)")

```

Fase 1 rediseñada: umbral de significancia (10) + estructura (T, F, B)

```

THRESHOLDS = [0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9]

# f_max calculado UNA SOLA VEZ aquí - se pasa a Fase 1 y Fase 2
f_max = sae_features_train.max(dim=0)[0] # (n_features,)
print(f"f_max calculado: {(f_max > 0).sum().item()} features activas de {f_max.shape[0]}")

start_time = time.time()

hp_mask_TFB, active_feature_indices = identify_high_precision_features_gpu(
    sae_features_train,
    bsp_gt_train_pieces,
    f_max,
    precision_threshold=0.95,
    significance_threshold=10,
    thresholds=THRESHOLDS,
    feature_batch_size=512,
    verbose=True
)

phase1_time = time.time() - start_time

print(f"\nTiempo Fase 1: {phase1_time:.2f} seg")
print(f"hp_mask_TFB shape: {hp_mask_TFB.shape} (T, n_active, n_bsps)")

```

```

f_max calculado: 8192 features activas de 8192
=====
FASE 1: IDENTIFICAR FEATURES DE ALTA PRECISIÓN
=====

```



```

Precision threshold:      0.95
Significance threshold:  >= 10 activaciones
Features activas:        8192/8192 (100.0%)
BSPs:                   128
Thresholds:              [0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9]
=====

```

Fase 1: 100%| | 16/16 [00:00<00:00, 255.83it/s]

Pares (feature, BSP) de alta precisión por threshold:

```

t=0.0:    0 pares
t=0.1:    5 pares
t=0.2:   53 pares
t=0.3:  424 pares
t=0.4: 1074 pares
t=0.5: 1183 pares ← más pares
t=0.6: 1115 pares
t=0.7: 1026 pares
t=0.8:  939 pares
t=0.9:  845 pares

```

Features HP por BSP en t=0.5:

```

Media:      9.2
Máx:       147
BSPs sin features: 66

```

Tiempo Fase 1: 0.87 seg

hp_mask_TFB shape: torch.Size([10, 8192, 128]) (T, n_active, n_bsps)

0.7 7. Fase 2: Reconstruir Tableros en Test Set (GPU)

```

def reconstruct_boards_gpu(
    sae_features_test,
    bsp_gt_test,
    hp_mask_TFB,
    active_feature_indices,
    f_max,
    thresholds=[0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9],
    verbose=True
):
    """
    Fase 2: barrido simultáneo de thresholds T → toma el mejor F1.

    Para cada threshold t:
    1. Binarizar features de test: active[p, f] = acts[p, f] > t * f_max[f]
    2. Predicción por (posición, BSP):
        predict[p, b] = OR_f ( active[p,f] AND hp_mask[t, f, b] )
        implementado como: (active_PF @ hp_FB) > 0
    3. F1 global micro-averaged (acumula TP/FP/FN sobre todas las posiciones y BSPs)

    Retorna el max F1 sobre todos los thresholds (Eq. 5 del paper: max_t).

    Args:
        sae_features_test: Tensor (P, n_features) GPU
        bsp_gt_test:      Tensor (P, n_bsps) GPU, bool
        hp_mask_TFB:      BoolTensor (T, n_active, n_bsps) de Fase 1
        active_feature_indices: LongTensor (n_active,) de Fase 1
        f_max:            Tensor (n_features,) GPU - MISMO objeto que Fase 1
        thresholds:       lista de fracciones (debe coincidir con Fase 1)
    """

```

```

Returns:
    best_f1:          float, max F1 sobre thresholds
    best_threshold:    float, threshold óptimo
    f1_per_threshold: np.array (T,), F1 por threshold
"""
device = sae_features_test.device
n_thresholds = len(thresholds)
thresholds_T = torch.tensor(thresholds, device=device)

# Activaciones del test solo para features activas
test_active_PF = sae_features_test[:, active_feature_indices] # (P, n_active)
active_f_max = f_max[active_feature_indices] # (n_active,)

gt_PB = bsp_gt_test.bool() # (P, B)
eps = 1e-8

f1_scores = torch.zeros(n_thresholds, device=device)
precision_T = torch.zeros(n_thresholds, device=device)
recall_T = torch.zeros(n_thresholds, device=device)

if verbose:
    print("=" * 60)
    print("FASE 2: RECONSTRUCCIÓN - BARRIDO DE THRESHOLDS")
    print("=" * 60)
    print(f"{'t':>5}  {'F1':>8}  {'P':>8}  {'R':>8}  {'TP':>8}  {'FP':>8}  {'FN':>8}")
    print("-" * 60)

for t_idx in range(n_thresholds):
    t = thresholds[t_idx]

    # Threshold absoluto por feature: t * f_max[f]
    abs_thresh_F = thresholds_T[t_idx] * active_f_max # (n_active,)

    # Binarizar features en test
    active_PF = test_active_PF > abs_thresh_F.unsqueeze(0) # (P, n_active)

    # predict[p, b] = any( active[p,f] AND hp[t,f,b] )
    # = (active_PF.float() @ hp_mask[t].float()) > 0
    hp_FB = hp_mask_TFB[t_idx].float() # (n_active, B)
    predictions_PB = (active_PF.float() @ hp_FB) > 0 # (P, B)

    # Métricas globales micro-averaged
    tp = (predictions_PB & gt_PB).sum().float()
    fp = (predictions_PB & ~gt_PB).sum().float()
    fn = (~predictions_PB & gt_PB).sum().float()

    p = tp / (tp + fp + eps)
    r = tp / (tp + fn + eps)
    f1 = 2 * p * r / (p + r + eps)

    f1_scores[t_idx] = f1
    precision_T[t_idx] = p
    recall_T[t_idx] = r

    if verbose:
        print(f"{'t':>5.1f}  {f1.item():>8.4f}  {p.item():>8.4f}  {r.item():>8.4f}"
              f"  {tp.item():>8.0f}  {fp.item():>8.0f}  {fn.item():>8.0f}")

best_idx = f1_scores.argmax().item()
best_f1 = f1_scores[best_idx].item()
best_threshold = thresholds[best_idx]

```

```

if verbose:
    print("=" * 60)
    print(f"Mejor threshold: t={best_threshold:.1f}")
    print(f"Reconstruction Score (max_t F1): {best_f1:.4f}")
    print("=" * 60)

    return best_f1, best_threshold, f1_scores.cpu().numpy()

print(" Fase 2 rediseñada: barrido T simultáneo + F1 global micro-averaged + max_t")

```

Fase 2 rediseñada: barrido T simultáneo + F1 global micro-averaged + max_t

```

start_time = time.time()

reconstruction_score, best_threshold, f1_per_threshold = reconstruct_boards_gpu(
    sae_features_test,
    bsp_gt_test_pieces,
    hp_mask_TFB,
    active_feature_indices,
    f_max,          # mismo tensor calculado antes de Fase 1
    thresholds=THRESHOLDS,
    verbose=True
)

phase2_time = time.time() - start_time
total_time = phase1_time + phase2_time

print(f"\nTiempo Fase 2: {phase2_time:.2f} seg")
print(f"Tiempo total: {total_time:.2f} seg ({total_time/60:.2f} min)")

```

```

=====
FASE 2: RECONSTRUCCIÓN - BARRIDO DE THRESHOLDS
=====

```

t	F1	P	R	TP	FP	FN
0.0	0.0000	0.0000	0.0000	0	0	2006000
0.1	0.0013	0.9889	0.0007	1335	15	2004665
0.2	0.0075	0.9658	0.0038	7539	267	1998461
0.3	0.0195	0.9614	0.0099	19762	793	1986238
0.4	0.0266	0.9685	0.0135	27011	878	1978989
0.5	0.0265	0.9790	0.0135	26985	579	1979015
0.6	0.0248	0.9828	0.0126	25229	442	1980771
0.7	0.0227	0.9876	0.0115	23078	289	1982922
0.8	0.0205	0.9907	0.0104	20819	195	1985181
0.9	0.0190	0.9977	0.0096	19278	45	1986722

```

=====
Mejor threshold: t=0.4
Reconstruction Score (max_t F1): 0.0266
=====

```

Tiempo Fase 2: 7.83 seg
Tiempo total: 8.71 seg (0.15 min)

0.8 8. Análisis de Resultados

```

print("=" * 60)
print("BARRIDO DE THRESHOLDS - F1 POR THRESHOLD")
print("=" * 60)
for i, (t, f1) in enumerate(zip(THRESHOLDS, f1_per_threshold)):

```

```

        marker = " ← MEJOR" if i == f1_per_threshold.argmax() else ""
        print(f"t={t:.1f}: F1={f1:.4f}-{marker}")
print("=" * 60)

plt.figure(figsize=(10, 6))
plt.plot(THRESHOLDS, f1_per_threshold, marker='o', linewidth=2, markersize=8)
plt.axhline(y=reconstruction_score, color='r', linestyle='--',
            label=f'Best F1={reconstruction_score:.4f} (t={best_threshold:.1f})')
plt.axhline(y=0.95, color='g', linestyle=':', label='Target Othello SAE (paper)=0.95')
plt.xlabel('Threshold t', fontsize=12)
plt.ylabel('F1 Score (micro-averaged global)', fontsize=12)
plt.title('Board Reconstruction: F1 por Threshold', fontsize=14, fontweight='bold')
plt.grid(True, alpha=0.3)
plt.legend()
plt.tight_layout()
plt.show()

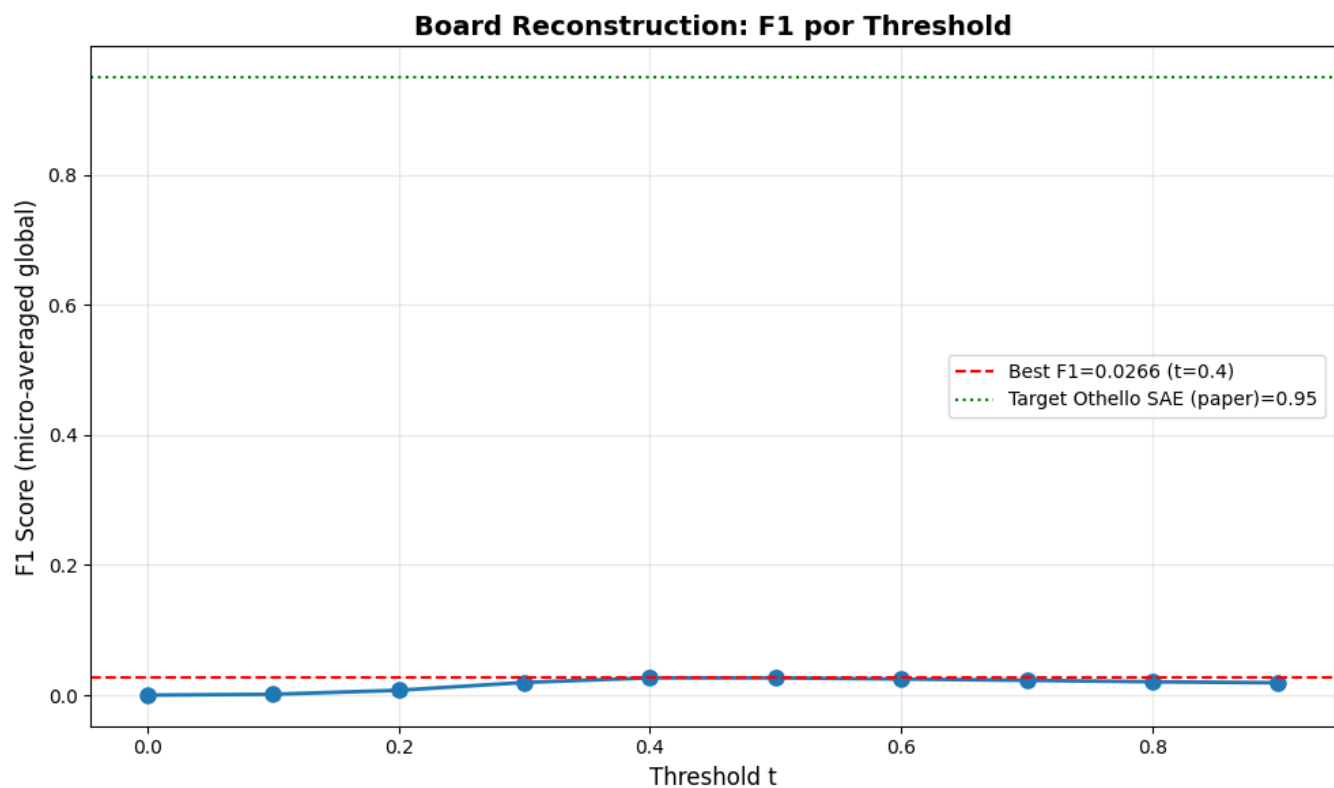
print(f"\n Threshold óptimo: t={best_threshold:.1f}")
print(f" Reconstruction Score: {reconstruction_score:.4f}")

```

```

=====
BARRIDO DE THRESHOLDS - F1 POR THRESHOLD
=====
t=0.0: F1=0.0000
t=0.1: F1=0.0013
t=0.2: F1=0.0075
t=0.3: F1=0.0195
t=0.4: F1=0.0266 ← MEJOR
t=0.5: F1=0.0265
t=0.6: F1=0.0248
t=0.7: F1=0.0227
t=0.8: F1=0.0205
t=0.9: F1=0.0190
=====

```



Threshold óptimo: t=0.4
Reconstruction Score: 0.0266

```
# Estadísticas sobre features de alta precisión por BSP
hp_per_bsp = hp_mask_TFB.sum(dim=1) # (T, n_bsps)
best_t_idx = torch.tensor(f1_per_threshold).argmax().item()
n_features_per_bsp = hp_per_bsp[best_t_idx].cpu().numpy()

print("="*60)
print("ESTADÍSTICAS - FEATURES POR BSP")
print("="*60)
print(f"Total BSPs: {len(n_features_per_bsp)}")
print(f"Features promedio por BSP: {np.mean(n_features_per_bsp):.1f}")
print(f"Features mediana por BSP: {np.median(n_features_per_bsp):.1f}")
print(f"BSPs con 0 features: {np.sum(n_features_per_bsp == 0)}")
print(f"BSPs con 1+ features: {np.sum(n_features_per_bsp > 0)}")
print(f"BSPs con 5+ features: {np.sum(n_features_per_bsp >= 5)}")
print(f"BSPs con 10+ features: {np.sum(n_features_per_bsp >= 10)}")
print(f"Max features en una BSP: {np.max(n_features_per_bsp)}")
print("="*60)
```

```
=====
ESTADÍSTICAS - FEATURES POR BSP
=====
Total BSPs: 128
Features promedio por BSP: 8.4
Features mediana por BSP: 1.0
BSPs con 0 features: 62
BSPs con 1+ features: 66
BSPs con 5+ features: 37
BSPs con 10+ features: 22
Max features en una BSP: 127
=====
```

```
# Comparación con resultados del paper
print("="*60)
print("COMPARACIÓN CON PAPER (Karvonen et al., NeurIPS 2024)")
print("="*60)
print(f"{'Modelo':<20} {'Reconstruction':<15}")
print("-"*60)
print(f"{'SAE random':<20} {'0.08':<15}")
print(f"{'SAE trained (paper)':<20} {'0.95':<15} ← Objetivo")
print(f"{'Linear probe':<20} {'0.99':<15}")
print(f"{'Nuestro SAE':<20} {reconstruction_score:<15.4f} ← Resultado")
print("="*60)

if reconstruction_score >= 0.90:
    print("\n Excelente: Muy cercano al objetivo del paper")
elif reconstruction_score >= 0.80:
    print("\n Bueno: Por encima de random pero debajo del objetivo")
else:
    print("\n Bajo: Revisar implementación o entrenamiento del SAE")
```

```
=====
COMPARACIÓN CON PAPER (Karvonen et al., NeurIPS 2024)
=====
Modelo                Reconstruction
-----
SAE random            0.08
SAE trained (paper)   0.95                ← Objetivo
Linear probe          0.99
Nuestro SAE           0.0266            ← Resultado
=====
```

Bajo: Revisar implementación o entrenamiento del SAE

0.9 9. Guardar Resultados

```
results = {
    'reconstruction_score': reconstruction_score,
    'best_threshold': best_threshold,
    'f1_per_threshold': f1_per_threshold,
    'thresholds': np.array(THRESHOLDS),
    'phase1_time_seconds': phase1_time,
    'phase2_time_seconds': phase2_time,
    'total_time_seconds': total_time,
    'bsp_names': bsp_pieces_names,
}

metrics_dir = get_metrics_dir(
    NAMING_META['experiment_base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    extras_id=extras_id
)
output_path = metrics_dir / f"reconstruction_{filter_suffix}.npz"
np.savez(output_path, **results)

print(f" Resultados guardados en: {output_path}")
print(f"\nContenido:")
print(f" reconstruction_score: {reconstruction_score:.4f}")
print(f" best_threshold:      {best_threshold:.1f}")
```

```

print(f"  fl_per_threshold:      {fl_per_threshold}")

# Comparación con el paper
print()
print("=" * 60)
print("COMPARACIÓN CON PAPER (Karvonen et al., NeurIPS 2024)")
print("=" * 60)
print(f"{'Modelo':<25} {'Reconstruction':>14}")
print("-" * 40)
print(f"{'SAE random':<25} {'0.08':>14}")
print(f"{'SAE trained (paper)':<25} {'0.95':>14} ← Objetivo Othello")
print(f"{'Linear probe':<25} {'0.99':>14}")
print(f"{'Nuestro SAE':<25} {reconstruction_score:>14.4f} ← Resultado")
print("=" * 60)

if reconstruction_score >= 0.90:
    print("\n Excelente: Muy cercano al objetivo del paper")
elif reconstruction_score >= 0.50:
    print("\n~ Moderado: Por encima de random, revisar SAE")
else:
    print("\n Bajo: Revisar entrenamiento del SAE o calidad de las activaciones")

```

Resultados guardados en: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_standard\sae_p-annealing_standard\12500games\ex3\metrics\reconstruction_absolute.npz

Contenido:

```

reconstruction_score: 0.0266
best_threshold:      0.4
fl_per_threshold:    [0.          0.00133011 0.00748731 0.01950305 0.02656094 0.02653961 0.02483572 0.02274404 0.0205
0.01903696]

```

```

=====
COMPARACIÓN CON PAPER (Karvonen et al., NeurIPS 2024)
=====
Modelo                Reconstruction
-----
SAE random              0.08
SAE trained (paper)     0.95 ← Objetivo Othello
Linear probe            0.99
Nuestro SAE             0.0266 ← Resultado
=====

```

Bajo: Revisar entrenamiento del SAE o calidad de las activaciones

```

import subprocess
from pathlib import Path
from sae.tools.experiment_utils import get_report_name

output_dir = get_metrics_dir(
    NAMING_META['experiment_base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    extras_id=extras_id
)

pdf_name = get_report_name(
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],

```

```

    f'reconstruction_{filter_suffix}',
    extras_id=extras_id
)

notebook_name = 'reconstruction_sae_standar2.ipynb'
notebook_abs = str(Path('.').resolve() / notebook_name)
output_path = output_dir / pdf_name

print(f' Generando PDF con Quarto...')
print(f' Archivo de salida: {output_path}')

result = subprocess.run(
    f'quarto render {notebook_abs} --to pdf --output-dir "{output_dir}" --output {pdf_name}',
    shell=True,
    capture_output=True,
    text=True
)

if result.returncode == 0:
    print(f' PDF generado exitosamente: {output_path}')
else:
    print(f' Error al generar PDF:')
    print(result.stderr)

```

Generando PDF con Quarto...

Archivo de salida: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_standard\sae_p-annealing_standard\12500games\ex3\metrics\sae_standard_p-annealing_l6_12500g_ex3_reconstruction_absolute.pdf
 PDF generado exitosamente: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_standard\sae_p-annealing_standard\12500games\ex3\metrics\sae_standard_p-annealing_l6_12500g_ex3_reconstruction_absolute.pdf

```

# =====
# SANITY CHECK: features sintéticas perfectas
# f_b[p] = 1.0 si BSP_b[p] = True, 0.0 si no.
# La métrica DEBE devolver 1.0. Si no, hay un bug.
# =====

import torch
import numpy as np
from pathlib import Path
from tqdm import tqdm

_root = Path('.././.././../').resolve()
_device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')

# --- Cargar BSP ground truth ---
_bsp_gt = np.load(_root / 'sae/metrics/02_data/bsp_ground_truth_2000games.npy') # cambia aquí
_bsp_names = np.load(
    _root / 'sae/metrics/02_data/bsp_ground_truth_2000games.names.npy',
    allow_pickle=True
)

# --- Split correcto 1000/1000 ---
_n_moves = 59
_n_games = 2000
_split_idx = (_n_games // 2) * _n_moves

_bsp_gt_train = _bsp_gt[:_split_idx]
_bsp_gt_test = _bsp_gt[_split_idx:]

# --- Filtrar BSPs de piezas ---
_piece_idx = np.array([
    i for i, n in enumerate(_bsp_names)
    if len(n) == 6 and n.startswith('BSP') and not n.endswith('0')
])

```



```

])
_train_pieces = torch.from_numpy(_bsp_gt_train[:, _piece_idx]).bool().to(_device)
_test_pieces = torch.from_numpy(_bsp_gt_test[:, _piece_idx]).bool().to(_device)

print(f"Device: {_device}")
print(f"BSPs de piezas: {len(_piece_idx)}")
print(f"Train: {_train_pieces.shape} | Test: {_test_pieces.shape}")

# --- Features sintéticas ---
_synth_train = _train_pieces.float()
_synth_test = _test_pieces.float()
_f_max_synth = _synth_train.max(dim=0)[0]

sparsity = (_synth_train == 0).float().mean()
n_active = (_f_max_synth > 0).sum().item()
print(f"Sparsity: {sparsity:.2%} | Features activas: {n_active}/{_f_max_synth.shape[0]}")

# --- Fase 1 ---
_THRESHOLDS = [0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9]
_thresh_T = torch.tensor(_THRESHOLDS, device=_device)
_n_T = len(_THRESHOLDS)
_act_idx = (_f_max_synth > 0).nonzero(as_tuple=True)[0]
_n_act = len(_act_idx)
_n_bsps = _train_pieces.shape[1]

_active_fmax = _f_max_synth[_act_idx]
_thresh_TF = _thresh_T[:, None] * _active_fmax
_on_count_TF = torch.zeros(_n_T, _n_act, device=_device)
_tp_TFB = torch.zeros(_n_T, _n_act, _n_bsps, device=_device)
_bsp_float = _train_pieces.float()
_batch = 128

print("\nFase 1 (sanity)...")
for fi in tqdm(range((_n_act + _batch - 1) // _batch)):
    fs, fe = fi * _batch, min((fi + 1) * _batch, _n_act)
    acts = _synth_train[:, _act_idx[fs:fe]]
    th_TF = _thresh_TF[:, fs:fe]
    active_TFP = acts.T.unsqueeze(0) > th_TF.unsqueeze(2)
    _on_count_TF[:, fs:fe] += active_TFP.sum(dim=2).float()
    _tp_TFB[:, fs:fe, :] += torch.einsum('tfp,pb->tfb', active_TFP.float(), _bsp_float)

_eps = 1e-8
_prec_TFB = _tp_TFB / (_on_count_TF.unsqueeze(2) + _eps)
_sig_mask = (_on_count_TF >= 1).unsqueeze(2)
_hp_mask_synth = (_prec_TFB >= 0.95) & _sig_mask

# --- Fase 2 ---
print("\nFase 2 (sanity)...")
_test_acts = _synth_test[:, _act_idx]
_gt_PB = _test_pieces.bool()
_fl_scores = torch.zeros(_n_T, device=_device)

print(f"{'t':>5} {'F1':>8} {'P':>8} {'R':>8}")
print("-" * 35)
for ti in range(_n_T):
    _abs_th = _thresh_T[ti] * _active_fmax
    _act_PF = _test_acts > _abs_th.unsqueeze(0)
    _pred_PB = (_act_PF.float() @ _hp_mask_synth[ti].float()) > 0
    tp = (_pred_PB & _gt_PB).sum().float()
    fp = (_pred_PB & ~_gt_PB).sum().float()
    fn = (~_pred_PB & _gt_PB).sum().float()
    p = tp / (tp + fp + _eps)

```

```

r = tp / (tp + fn + _eps)
f1 = 2 * p * r / (p + r + _eps)
_f1_scores[ti] = f1
print(f"{_THRESHOLDS[ti]:>5.1f}  {f1.item():>8.4f}  {p.item():>8.4f}  {r.item():>8.4f}")

_score_synth = _f1_scores.max().item()
_best_t      = _THRESHOLDS[_f1_scores.argmax().item()]

print()
print("=" * 50)
print("RESULTADO SANITY CHECK")
print("=" * 50)
print(f"Reconstruction Score (sintético): {_score_synth:.4f}")
print(f"Mejor threshold: t={_best_t:.1f}")
if abs(_score_synth - 1.0) < 0.01:
    print(" SANITY CHECK PASADO: score 1.0 → la métrica está bien implementada.")
else:
    print(f" SANITY CHECK FALLIDO: se esperaba 1.0, se obtuvo {_score_synth:.4f}")
    print(" Revisar Fase 1 / Fase 2.")
print("=" * 50)

```

Device: cuda
 BSPs de piezas: 256
 Train: torch.Size([59000, 256]) | Test: torch.Size([59000, 256])
 Sparsity: 73.44% | Features activas: 256/256

Fase 1 (sanity)...

100%| | 2/2 [00:00<00:00, 1209.43it/s]

Fase 2 (sanity)...

t	F1	P	R
0.0	1.0000	1.0000	1.0000
0.1	1.0000	1.0000	1.0000
0.2	1.0000	1.0000	1.0000
0.3	1.0000	1.0000	1.0000
0.4	1.0000	1.0000	1.0000
0.5	1.0000	1.0000	1.0000
0.6	1.0000	1.0000	1.0000
0.7	1.0000	1.0000	1.0000
0.8	1.0000	1.0000	1.0000
0.9	1.0000	1.0000	1.0000

```

=====
RESULTADO SANITY CHECK
=====
Reconstruction Score (sintético): 1.0000
Mejor threshold: t=0.0
SANITY CHECK PASADO: score 1.0 → la métrica está bien implementada.
=====

```